

Evaluating Prediction Quality from Decision Outcomes: A Novel Data-Driven Framework for EVB Supply Chain Networks with Winner Determination-Based Third-Party Supplier Integration

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ARTICLE INFO

Keywords:

Electric vehicle

Network design

Data-driven

Winner determination

ABSTRACT

As the core component of electric vehicles (EVs), the stable supply of EV batteries (EVBs) is a decisive constraint on EV market expansion, and the EVB production-or-procurement decision is a key strategic concern for manufacturers. This study optimizes the EVB manufacturing supply chain network design (EVBMSCND) under a novel data-driven framework, based on an equilibrium perspective on EVB production and procurement. The primary contributions of this paper are summarized in two aspects. First, the winner determination (WD) method for combinatorial auctions is innovatively incorporated into EVBMSCND to integrate third-party EVB suppliers, thereby realizing a production–procurement equilibrium. Second, a novel data-driven optimization framework is proposed, in which prediction models are evaluated based on the quality of the resulting network decisions rather than predictive accuracy, thus achieving an integrated coupling of the prediction and optimization stages. To address this problem, a two-stage multi-period stochastic programming model is formulated and subsequently reformulated as a mixed-integer linear programming model. A scenario relaxation and decomposition-based approximation algorithm is developed. Finally, the effectiveness of the proposed model and algorithm is verified through a case study generated from data from a real-world EV enterprise. The results prove that the proposed integrated WD framework effectively enhances the economic performance and responsiveness of EVBMSCND under demand uncertainty. Furthermore, the decision-oriented evaluation mechanism for prediction models is shown to guide high-quality strategic decisions more effectively than traditional methods that rely exclusively on predictive accuracy.

1. Introduction

Driven by the global emphasis on environmental protection and sustainable development, rapid expansion is observed in the electric vehicle (EV) industry. Global new EV sales are recorded at over 17 million units, and the total global stock is maintained at nearly 58 million units, as reported by the International Energy Agency¹. Higher efficiency and flexibility in EV manufacturing supply chains are necessitated by market requirements for capacity expansion and delivery cycles. Electric vehicle battery (EVB), as the core component of EV, directly determine EV pricing competitiveness and manufacturers' profit margins, thereby becoming a key bottleneck that constrains the efficient and flexible development of EV manufacturing supply chains. To break the dual dilemma of "cost passivity" and "supply chain dependency" imposed by EVB manufacturers, battery self-production networks are established by EV manufacturers. For instance, strategic initiatives in battery research and development were actively pursued by Tesla starting in 2019². Notably, the superior cost-effectiveness of the self-produced 4680 battery production line

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¹International Energy Agency (IEA). Global EV Outlook 2024: Trends in electric vehicle sales and stock. Available at: https://auto.cnr.cn/2015xc/20250519/t20250519_527175308.shtml (accessed 25 February 2026).

²Tesla. Strategic initiatives in battery research and development. Available at: <https://caifuha0.eastmoney.com/news/20251113162920691342620> (accessed 24 February 2026).

compared to external suppliers was announced during an internal meeting in the first quarter of 2025³. Similarly, a self-production rate of 95% for the Blade Battery was achieved by BYD, as demonstrated in the 2024 Sustainability Report, whereby a “vehicle–battery” integrated supply chain is established⁴. Furthermore, the production domain is penetrated by traditional automakers, such as Volkswagen and BMW, through mechanisms including joint ventures with Northvolt and the independent development of solid-state batteries⁵. A “self-built” mode is thus adopted by these entities for the mitigation of the dual constraints of cost and supply chain dependencies.

Network design (ND), as a critical stage that determines operational processes, facility location, and resource allocation, plays a crucial role in the efficient development of EVB manufacturing supply chain (EVBMSC). The critical role of network optimization in enhancing the efficiency of EVBs supply chains is emphasized in recent studies (Chen et al., 2023; Huang et al., 2024). However, location decisions for critical facilities such as EVBs production plants are primarily concentrated on by existing studies, whereas the systemic optimization of network topology and flows is relatively neglected. A perspective on enhancing supply chain efficiency by integrating complementary resources is offered by the emergence of fourth-party logistics (4PL). Addressing the limitation of isolated decision-making, the achievement of superior economic benefits is demonstrated by Yin et al. (2024) by integrating facility location and transportation link selection decisions to optimize network topology, rather than merely increasing the number of facilities.

Uncertainty in EVs market demand is driven by raw-material supply constraints, policy changes, and rapid technology evolution. Demand is regarded as a critical parameter that drives network operations and directly influences network optimization results. How to enhance network flexibility to accommodate demand fluctuations is one of the core concerns of EV manufacturers. Accurate demand forecasting effectively mitigates risks and improves network flexibility. In recent years, time-series forecasting is supported by the application of neural network models such as gated recurrent unit (GRU), Transformer, and long short-term memory (LSTM), and data-driven optimization methods are investigated by integrating parameter forecasting with network optimization. For example, a data-driven two-stage stochastic programming model is proposed by Fattahi (2020), in which uncertain parameters are estimated from historical data and network configuration is optimized, demonstrating the effectiveness of data-driven approaches in supply chain network design. However, conventional forecasting methods have limitations, mainly because model performance is evaluated solely by forecasting accuracy, while such accuracy metrics often fail to effectively reflect the quality of optimization decisions. For EVBMSC network design, forecasting model performance is reflected in whether superior optimization decisions are induced. Therefore, when optimization decision results are used as the evaluation criterion for forecasting methods, reliance on forecasting accuracy while neglecting decision effectiveness is avoided. Based on this rationale, a new data-driven optimization framework is proposed in this study, in which forecasting models are evaluated by optimization decision results, thereby enabling an effective integration of forecasting and optimization decision-making.

A further challenge presented by the uncertainty of EVs market demand is associated with the indeterminacy of the EVBs manufacturing network scale. Consequently, network design is deemed a pivotal decision problem at the strategic level of the EVBs supply chain, wherein the network is configured before demand is realized. On the one hand, excessive reliance on self-built networks is prone to inducing battery overcapacity and the idling of fixed assets during demand downturns, thereby escalating operational costs. On the other hand, if network capability is insufficient, reliance on third-party suppliers (TPSs) is necessitated, and the issue of cost containment remains unresolved. Therefore, a balanced network integrating production and procurement is constructed to ensure the stability and flexibility of the EVBMSC, while cost risks are effectively mitigated. However, in recent years, a rapid expansion in the number of EVB manufacturers is driven by the widespread adoption of EVs. As a result, supplier selection among numerous EVB suppliers is regarded as a critical problem. In this context, auctions are widely employed as efficient resource allocation mechanisms in complex transaction processes within market economies. Combinatorial auctions are enabled to achieve optimal resource allocation by constructing the winner determination problem (WDP), providing a new approach for EVB procurement. In this study, winner determination (WD) method is innovatively integrated into the

³Tesla. Announcement on 4680 battery production line cost-effectiveness. Available at: <https://tech.huanqiu.com/article/4MB8xbdsbgD> (accessed 25 February 2026).

⁴BYD. BYD 2024 Sustainability Report and Blade Battery production analysis. Available at: <https://data.eastmoney.com/notices/detail/002594/AN202503241646931773> (accessed 24 February 2026).

⁵Volkswagen. Joint ventures with Northvolt and solid-state battery development. Available at: <https://www.jiemian.com/article/3217527.html> (accessed 25 February 2026).

EVBMSC network design problem. By introducing TPSs through WD, network flexibility is enhanced, and the most cost-effective balance between EVBs production and external procurement is achieved.

In this study, a novel data-driven multi-period EVBMSC network design problem is investigated, wherein third-party EVBs suppliers are integrated based on WD to enhance flexibility. Three models, namely GRU, Transformer, LSTM and are utilized to predict the fluctuation characteristics of demand. Subsequently, a novel data-driven optimization framework is constructed, in which the optimization results are employed to evaluate and select the forecasting methods. The WD method is innovatively integrated into EVBMSC network design to incorporate third-party EVBs suppliers, thereby enhancing network flexibility to mitigate operational risks induced by demand volatility. A two-stage multi-period stochastic EVBMSC network design model is constructed, which is further formulated as a Mixed-Integer Linear Programming (MILP) model. A Scenario Relaxation and Decomposition-based Approximation Algorithm is proposed to tackle the solution complexity arising from sampling, characterized by the integration of Sample Average Approximation (SAA) and Latin Hypercube Sampling (LHS) with Dual Decomposition and Lagrangian Relaxation (LR) methods. A numerical instance based on real-world EVs manufacturing data is generated to verify the performance of the proposed modeling and solution approach. Operational efficiency and responsiveness are significantly enhanced by the WD-based integration of third-party suppliers. Moreover, by prioritizing decision outcomes over prediction accuracy, evaluation criteria are tailored to reflect actual application value, and the potential imbalance between forecasting performance and decision efficacy is effectively avoided. The main contributions of this paper are summarized as follows:

1. While data-driven optimization frameworks integrating prediction and decision-making have been widely investigated, limited attention has been paid to evaluating forecasting models from the perspective of downstream optimization outcomes. To address this limitation, this study proposes a novel data-driven optimization framework and develops a forecasting model evaluation mechanism grounded in optimization results, thereby establishing a dynamic feedback loop between demand forecasting and optimization decision-making.
2. The WD approach is incorporated into the EVBMSC network to coordinate TPSs, enabling the network to respond more flexibly to demand uncertainty
3. A data-driven two-stage stochastic multi-period EVBMSC network design model incorporating the WD method is developed. To solve the resulting problem efficiently, an approximate algorithm based on scenario relaxation and decomposition is further proposed.

The organization of the remaining sections is outlined below. In Section 2, literature concerning EVBMSC network design, WDP, and data-driven forecasting is systematically reviewed. The problem description and corresponding mathematical formulation are presented in Section 3. In Section 4, a novel Scenario Relaxation and Decomposition-based Approximation Algorithm is proposed. The effectiveness of the model and the algorithm is validated through numerical experiments in Section 5, where an analysis of key parameters is also conducted. Finally, the main work is summarized, key contributions are listed, and future research directions are discussed in Section 6.

2. Literature Review

In this study, a data-driven EVBMSC network design problem is investigated, integrating WD with optimization and time-series forecasting. Accordingly, the relevant literature across these interdisciplinary domains is reviewed in this section.

2.1. EVBMSC Network Design

Driven by the expanding scale of the EVs market, the design of the EVBMSC network is highlighted. A relatively early study was conducted by Kalaitzi et al. (2019) focusing on micro-level relationships within the EVB manufacturing network, wherein network design is characterized not as a static physical connection, and the role of core suppliers as “coordinators” is necessitated by effective design strategies. Subsequently, the scope of EVB supply chain network design was extended to a macro-regional level by Li et al. (2025). The core design focus of modern EVB manufacturing networks is revealed to be shifted towards the integration of comparative advantages across cross-national regions, aimed at the establishment of a specialized division of labor system. In recent years, confronted with the intensification of global geopolitical tensions, the focal point of network design is shifted towards resilience. The establishment process of the EVB supply chain was analyzed from a dynamic evolutionary perspective by Chen et al. (2023), wherein collaborative innovation and information sharing modules are advocated as indispensable components to form a soft

defense mechanism against geopolitical conflicts. Building upon this foundation, a quantitative reconstruction of the topological structure of the EVB manufacturing network was conducted by Huang et al. (2024) utilizing complex network theory. Security vulnerabilities within the prevailing scale-free network structure are identified, and the capability of the EVBMS network to withstand targeted attacks is enhanced by the proposed reconstruction.

Although the EVBMS network design problem was investigated by previous scholars, location decisions for key facilities, such as battery production plants, are prioritized in the majority of existing studies, while the optimization of the overall network topological structure is neglected. The 4PL is identified by Yin et al. (2024) to integrate complementary resources from multiple third-party logistics providers (3PLs), facilitating collaborative decision-making on facility location and transportation supplier selection for the optimization of network topological structure and the consequent enhancement of operational efficiency. Currently, the supply chain network design (SCND) problem is investigated from the 4PL perspective in existing literature. Customer satisfaction was incorporated into the fourth-party logistics network design (4PLND) problem under a deterministic environment in a relatively early study conducted by Huang et al. (2021). The scope of the investigation was subsequently expanded by Wang et al. (2021) through the integration of both supply and demand satisfaction into the problem. However, customers are primarily characterized as static receivers in prior research, with the active influence of the customer side being neglected. In fact, the customer side is identified not merely as the terminal point of the manufacturing supply chain (MSC), but as a primary driver for network operation and a critical criterion for the assessment of service levels. Research on service-oriented MSC was initially conducted by Zhang et al. (2024) focusing on customer behavioral characteristics, wherein service time and delivery quantity were incorporated as two independent service-level dimensions for the investigation of assessment period impacts. Subsequently, the prior investigation into customer behavior was expanded by Zhang et al. (2025a) by integrating a demand fluctuation mechanism driven by customer satisfaction into network design, and the profit deviation induced by the neglect of customer psychological factors was underscored. Furthermore, the research scope of psychological behavior was extended from customers to decision-makers themselves by Zhang et al. (2026) through the introduction of the promised service time concept, wherein regret behavior arising from the deviation between promised and actual performance was incorporated into the model to facilitate the trade-off between service punctuality and profit. In addition to demand-side fluctuations induced by customer behavior, wide-ranging uncertainty challenges originating from the supply side and logistics segments are confronted by MSCs in practical operations. The dual disruption risks of facilities and 3PLs were incorporated into ND optimization under uncertain environments in a relatively early study conducted by Yin et al. (2022). Subsequently, a temporary outsourcing service strategy was proposed by Yin et al. (2024) in response to uncertainty-induced demand spillover issues, wherein a dynamic mechanism for addressing emergency demand was integrated into ND for the enhancement of supply chain performance. Recently, the ND problem was further deepened from the viability perspective by Zhang et al. (2025b), wherein agility, resilience, and survival sustainability were simultaneously incorporated into the consideration for the reinforcement of network defense and recovery during a mega-crisis. The consideration of the long-term industrial development environment, particularly within the context of global carbon neutrality, is necessitated once the capability of supply chains to address uncertainty risks is established. Carbon cap-and-trade policies were incorporated into 4PLND by Zhang et al. (2025c), wherein the trade-off between costs and carbon emissions under service time constraints was quantified in detail for the collaborative optimization of economic benefits and environmental sustainability.

A gradual increase in the volume of research concerning 4PLND is identified based on the preceding literature review, yet analysis pertaining to the impact of 4PL concepts on the EVBMS network design problem is found to be scarce. The EVBMS network design problem is investigated from a 4PL perspective in the present study, whereby theoretical support and managerial insights for the enhancement of supply chain competitiveness through the 4PL model are provided.

2.2. Winner Determination Problem

Combinatorial Auctions are characterized by De Vries and Vohra (2003) as a market mechanism wherein the submission of package bids on item combinations is permitted, whereby value interdependencies between items are effectively revealed and optimal resource allocation is achieved. WDP is identified as the core challenge of combinatorial auctions, fundamentally characterized as an NP-hard set packing problem. A vast body of literature was generated in this domain, driven by substantial application value (Rothkopf et al., 1998; Sandholm, 2002). Initial WDP research was centered on the efficient allocation of scarce public resources, whereby the status of WDP as

an effective tool for large-scale resource allocation was established through pioneering applications in airport slot allocation (Rassenti et al., 1982).

With the emergence of the 4PL mode, the application of WDP is brought to the forefront of supply chain manufacturing, wherein complex global transportation services are procured from numerous third-party logistics (3PL) providers by service integrators through combinatorial reverse auctions (Huang et al., 2015; Hu et al., 2016). Deterministic models are predominantly utilized for WDPs, whereas significant uncertainty is encountered in real-world scenarios. To address fluctuations in customer freight volumes, stochastic programming (Ma et al., 2010) and two-stage robust optimization methods (Remli and Rekik, 2013; Zhang et al., 2015) were introduced, whereby robust solutions under demand uncertainty are provided. Supply-side uncertainty is also identified, whereby the dual uncertainty of freight volume and carrier capacity was incorporated into further expanded models by Remli et al. (2019). A two-stage sample-robust WDP model under budget constraints was proposed by Li et al. (2023), wherein cost overrun expectations and tail risks are significantly reduced through the introduction of a risk index indicator, providing new perspectives for cost control in transportation procurement. With the shift of the supply chain manufacturing focus towards sustainability, flexibility, and resilience, a continuous expansion in the application scope of WDP is observed. For instance, in response to the call for sustainable development, sustainability attributes of suppliers were incorporated into WDP models by Qian et al. (2021). Sustainability goals are integrated into procurement auctions, whereby cost reduction and efficiency enhancement are realized through the introduction of carbon emission reduction performance and discount bidding mechanisms (Triki et al., 2025). For the enhancement of response capability to market risks, logistics service flexibility was quantified and incorporated into decision-making by Khakdaman et al. (2022). Furthermore, complex factors including disruption risks and quantity discounts were also integrated into WDP models, aimed at the construction of more flexible procurement strategies (Qian et al., 2020; Yin et al., 2021). With the development of research, the research perspective of WDP is gradually expanded from singular resource allocation towards a comprehensive integration with complex optimization problems. In the downstream segments of logistics and supply chains, the direct embedding of operations research optimization models, such as the Vehicle Routing Problem (VRP), into the WDP decision-making framework is attempted for the resolution of increasingly complex combinatorial optimization challenges. For instance, within emergency rescue scenarios, the efficiency of task allocation and route scheduling was enhanced through the improvement of the WDP algorithm by Abusalama et al. (2022). In the field of crowdsourced logistics, WDP is characterized not merely as price-driven competition but is transformed into a route optimization problem incorporating spatio-temporal constraints. For the new retail scenario, blockchain technology was introduced by Li et al. (2024), whereby the collaborative scheduling of crowdsourcing and self-operated delivery was resolved through a two-tier allocation mechanism.

Based on the preceding literature review, the embedding of WD methods into optimization problems such as VRP is identified for the achievement of a dynamic balance between long-term resource holding and short-term procurement. Although a theoretical foundation for the present research is established by the prior literature, the integration of WD into ND is found to be scarce.

2.3. Time-series Forecasting and Data-driven Optimization

Demand forecasting (DF) is identified as a core component of modern supply chain design, and the efficiency of the entire supply chain is directly influenced by the quality of DF (Sasser, 1976; Chandra and Fisher, 1994). Moreover, the foundation for all subsequent decisions is established by DF (Haberleitner et al., 2010). Traditional DF is primarily based on classical linear models. Owing to their strong capability to explain linear trends, the Auto-Regressive Integrated Moving Average (ARIMA) model and its seasonal variants were widely applied in power load forecasting (Wang et al., 2012), while linear regression models were used to forecast national electricity consumption in Italy (Bianco et al., 2013). Traditional econometric models are primarily constrained by the linear assumption, whereby the accurate capturing of complex variation patterns is hindered by nonlinear dynamic relationships prevalent in the real world (Gao et al., 2023b).

To overcome the limitations of traditional methods, attention was gradually directed to machine learning (ML). Artificial Neural Networks (ANNs), characterized by strong nonlinear modeling capabilities (Perrusquía and Yu, 2021; Gao et al., 2023a), exhibit superior performance in processing complex data patterns, and traditional models such as linear regression are outperformed (Wu et al., 2020). Within comparative research on retail sales forecasting, significant predictive effectiveness was exhibited by ANNs, while traditional methods were surpassed (Alon et al., 2001). Various hybrid forecasting models were developed through the integration of traditional models and artificial intelligence (AI) technologies. The synergy of distinct methodological advantages is targeted by hybrid models. For instance,

the linear modeling capability of ARIMA was integrated with the nonlinear mapping capability of neural networks by Valenzuela et al. (2008). In addition, linear components within time series were processed by ARIMA, followed by the implementation of nonlinear modeling, through which superior performance in tasks such as oil well production forecasting was achieved by Fan et al. (2021). With the rapid advancement of deep learning technologies, Recurrent Neural Networks (RNNs) and variants, including LSTM and GRU, are established as research hotspots in demand forecasting, attributed to advantages in the processing of sequential data. Customer demand is fundamentally modeled as time series data (Villegas et al., 2018; Abbasimehr et al., 2020), whereby time dependency is effectively addressed by RNNs through an internal feedback loop structure (Chandriah and Naraganahalli, 2021). However, traditional RNNs are constrained by gradient vanishing and short-term memory during long sequence processing, whereby the capturing of long-term dependencies is hindered and prediction accuracy is adversely impacted (Sagheer and Kotb, 2019).

LSTM was successfully utilized to forecast sales performance within complex supply chains (Weng et al., 2019), customer demand in furniture enterprises (Abbasimehr et al., 2020), and daily sales volumes of fresh fish (Marques da Fonseca, 2020). Concurrently, GRU is favored for comparable performance with LSTM and a simplified architecture, whereby wide application in domains such as apparel sales forecasting is observed (Li et al., 2021). Recently, Transformer models are increasingly utilized in the field of time series forecasting. The parallel processing of entire sequences and the direct capturing of dependencies between any two positions through the self-attention mechanism are enabled, whereby significant potential for the modeling of long-term dependencies is demonstrated (Park et al., 2023).

DF is frequently integrated with data-driven approaches. As a paradigm that learns from data and guides decision-making, data-driven approaches were long regarded as an important pathway in scientific research (Langley and Zytkow, 1989), and their theoretical foundations were grounded in the seminal work on pattern recognition and machine learning (Bishop, 2006).

In the context of supply chain complexity, the utilization of data-driven methodologies to mitigate environmental uncertainty is prioritized as a primary research objective. The necessity for an enhanced representational capacity of models regarding uncertainty was established in early studies (Reichert and Vanrolleghem, 2001). Accordingly, Bayesian inference was applied (Borsuk et al., 2002) to update model parameter posterior distributions via observational data. Extensive simulated datasets were generated through the integration of Monte Carlo simulations, and more accurate risk assessments for decision-making were achieved. As research progressed, challenges in practical data acquisition, such as incomplete information, class imbalance, and data noise, were prioritized. The maintenance of model reliability and generalization capability under low-quality data conditions was investigated (Wang and Yao, 2012; Liu et al., 2015). Beyond uncertainty, the enhancement of supply chain architecture flexibility and responsiveness via data-driven methodologies was also prioritized. Within the research of Chiu and Lin (2004), the integration of technologies such as ANN was demonstrated to facilitate collaborative supply chain planning. Such efforts represented an early attempt at integrated intelligent systems. At tactical and operational levels, dynamic optimization via real-time data is emphasized within data-driven applications. For instance, an online control framework based on real-time passenger and vehicle data was recently established (Kolcheva et al., 2025). Significant improvements in operational efficiency were achieved through the implementation of dynamic adjustment strategies. With the advancement of applications, data-driven models are integrated with broader supply chain objectives. For instance, within the context of the circular economy, data-driven methodologies were utilized to optimize recycling center locations (Yahyapour Ganji et al., 2025). Consequently, a closed-loop network characterized by both flexibility and responsiveness was established. Open innovation and blockchain technology were adopted as strategies to enhance supply chain anti-fragility (Lotfi et al., 2024). Effectiveness in cost reduction was validated via a data-driven robust optimization model. Furthermore, a dual-channel closed-loop supply chain was investigated (Gao et al., 2024), where data-driven techniques were utilized to characterize uncertain market demand. Data-driven methodologies are extensively integrated across multiple levels of supply chain manufacturing.

Although data-driven optimization frameworks combining prediction and decision-making are investigated, the evaluation of prediction effectiveness via optimization decision outcomes is scarcely explored.

2.4. Research Gap

While advances are achieved in data-driven optimization, WDP, and 4PL, the integration of these domains to address complex challenges within the EVBMS is scarcely investigated.

- Existing studies commonly treat prediction and decision-making as separate tasks, which constitutes a fundamental limitation in data-driven decision systems. In most cases, forecasting models are evaluated primarily based

on predictive accuracy, whereas their ultimate role in supporting decision-making processes is often overlooked. Consequently, how forecasting models should be evaluated and selected from the perspective of final decision quality remains insufficiently explored.

- Research on WD methods and 4PL network design has largely been conducted independently. The integration of WD into multi-period EVBMS network design, while incorporating third-party EVB suppliers through 4PL to enhance network flexibility and robustness under market fluctuations, remains underexplored.

In response to the above limitations, a data-driven multi-period EVBMS network design problem, characterized by the embedding of DF into WD, is addressed from a 4PL perspective. Prediction models are innovatively evaluated via decision results. Three distinct time-series forecasting models are employed for future demand prediction. Subsequently, a data-driven two-stage stochastic programming model is constructed based on prediction results, whereby the collaborative optimization of logistics service procurement and supply network structure is achieved. Through the integration of 4PLND, data-driven, WD, and time-series forecasting techniques, the flexibility, service level, and decision-making intelligence of the multi-period EVBMS system are enhanced.

3. Problem Description and Model Formulation

In this section, time series forecasting methods are described, after which the multi-period EVBMS network design problem is elaborated and the corresponding model formulations are presented.

3.1. Problem Description

In this study, DF is regarded as the cornerstone of the data-driven decision-making framework, where prediction accuracy directly influences the effectiveness of subsequent ND. Accordingly, public historical demand data from a new energy vehicle manufacturer were utilized, and three representative deep learning models were employed to predict and compare demand distribution characteristics. LSTM (Hochreiter and Schmidhuber, 1997) addresses the long-term dependency issue of RNNs through the introduction of a cell state and three gating units: the forget, input, and output gates. Information is selectively retained or forgotten via this structure, whereby key signals are effectively transmitted across long sequences and gradient vanishing is avoided, as illustrated in Subplot c of Figure 1c. GRU is recognized as a variant where the long-sequence gradient problem of RNNs is addressed via reset and update gates (Cho et al., 2014). New states are generated by controlling the flow of historical information and current inputs, as depicted in Subplot a of Figure 1a. Due to a simpler structure, fewer parameters, and performance comparable to LSTM, GRU is widely favored. The Transformer model was proposed by Vaswani et al. (2017) in “Attention Is All You Need”. Recurrence and convolution structures extensively used in traditional sequence processing tasks are completely discarded in this model. Instead, dependence is placed entirely on a mechanism named Self-Attention for the capture of intra-sequence dependencies, as shown in Subplot b of Figure 1b.

Significant differences in core mechanisms are exhibited by these three mainstream time-series models. Information is processed sequentially step-by-step by LSTM and GRU, as RNNs, utilizing gating mechanisms to capture temporal dependencies. Conversely, recurrence structures are discarded in Transformer, where intra-sequence associations at all positions are computed in parallel via the self-attention mechanism, whereby a natural advantage in capturing ultra-long-term dependencies is possessed. Despite distinct implementation paths, the ultimate goal of learning complex non-linear temporal patterns to provide high-precision prediction inputs for optimization decisions is shared by all three models. Through the comprehensive evaluation of these prediction models from different paradigms, the selection of the most reliable data support for downstream multi-period EVBMS network optimization is aimed at in this study, whereby the robustness and effectiveness of the entire decision-making framework are enhanced.

Regarding EVBMS network design, an EVBMS network is abstractly constructed based on the actual factory layout of the new energy vehicle manufacturer, where a potential 4PL management framework is revealed, as shown in Figure 2. For the multi-period EVBMS network design problem, demand, costs, and capacities are all correlated with periods, and significant time dependencies are exhibited. The demand side of the network is constituted by multiple EVB manufacturing plants, denoted by black dots in the figure, where a specific battery demand quantity in each period and a unit penalty cost incurred when demand is unsatisfied are associated with each plant. To satisfy plant production requirements, two EVB supply sources are identified. The first source consists of potential self-built battery production plants, denoted by yellow dots in the figure, where capacity limits, fixed costs, and unit production costs are associated with each plant in every period, and activation is determined by a binary variable. The second source comprises external

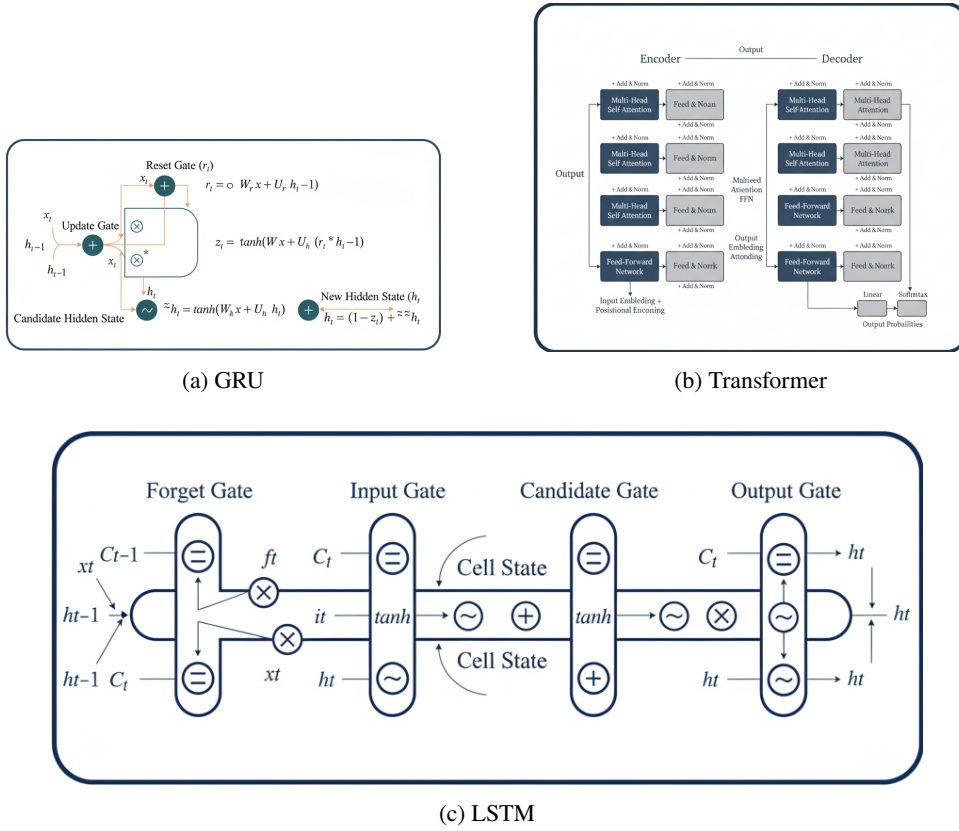


Figure 1: Predictive model principles

power battery suppliers, where multiple packages can be submitted by each external EVB supplier in every period. Capacity upper bounds, fixed costs, and unit costs are possessed by each package, and selection is similarly controlled by a binary variable. Logistics tasks connecting supply nodes and demand nodes are outsourced to 3PL suppliers. On each potential transportation route, multiple distinct 3PL suppliers are available for selection in every period, where independent capacity limits, fixed costs, and unit transportation costs are possessed by each supplier, and activation is also reflected by a binary decision variable.

With reference to the data-driven rolling horizon optimization framework proposed by Fattahi and Govindan (2022), the prediction component within the data-driven multi-period EVBMS network design decision framework can be briefly described as Figure 3. At each decision point t , the characteristic values of the random distribution of demand for the future 3 periods are predicted based on data from the past 12 periods. Upon execution of the prediction, data is updated with the results, and the entire planning horizon is advanced by three periods. At the subsequent decision point $t + 3$, the optimization process is repeated. Data-driven management of uncertain optimization is achieved by this method through continuous “replanning” and “feedback correction”.

To formulate the required mathematical model, relevant notations are defined and summarized in Table 1.

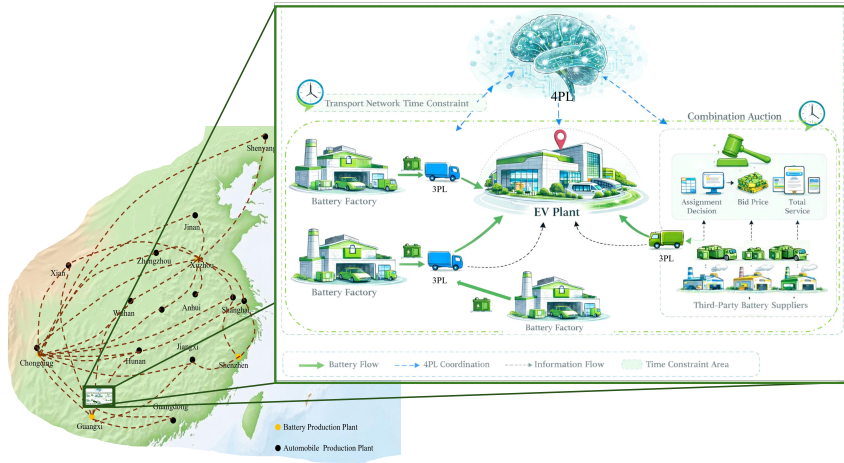


Figure 2: EVBMS network

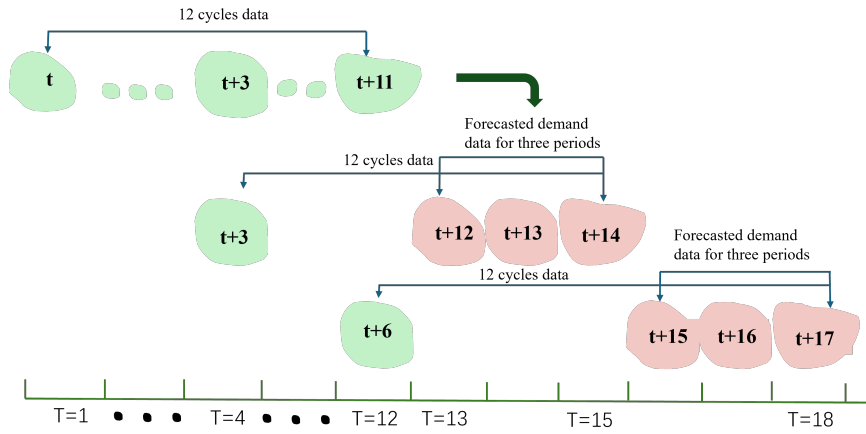


Figure 3: Principle of prediction

Table 1
Symbol description

Symbol	Description
Sets	
$\mathcal{T}$	Set of time periods.
B	Set of battery production plants.
E	Set of car manufacturing plants.
O	Set of battery suppliers.
L_{be}	Set of 3PL providers for the route from battery plant b to car manufacturing plant e , for each $b \in B, e \in E$.
K_o	Set of package options offered by supplier o , for each $o \in O$.
Ξ	Set of scenarios, indexed by ς , for each $\varsigma \in \Xi$.
H_{bt}	Fixed cost of operating battery plant b in period t , for each $b \in B, t \in \mathcal{T}$.
v_{ot}	Maintenance cost for supplier o in period t , for each $o \in O, t \in \mathcal{T}$.
h_{belt}	Maintenance cost for 3PL provider l on the route from b to e in period t , for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
sc_{bt}	Unit production cost at battery plant b in period t , for each $b \in B, t \in \mathcal{T}$.
kc_{belt}	Unit transportation cost via 3PL provider l from b to e in period t , for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
p_{et}	Penalty cost per unit of unmet demand at car manufacturing plant e in period t , for each $e \in E, t \in \mathcal{T}$.
b_{okt}	Procurement cost of package k from supplier o in period t , for each $o \in O, k \in K_o, t \in \mathcal{T}$.
$N_t^{\min}$	Minimum total quantity in period t , for each $t \in \mathcal{T}$.
$N_t^{\max}$	Maximum total quantity in period t , for each $t \in \mathcal{T}$.
τ_{belt}	Transportation time via 3PL provider l from b to e in period t , for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
v_{eokt}	Transportation time for package k from supplier o to car plant e in period t , for each $e \in E, o \in O, k \in K_o, t \in \mathcal{T}$.
κ_t	Maximum allowable time limit in period t , for each $t \in \mathcal{T}$.
Λ	Maximum inventory capacity.
Sq_{bt}	Production capacity of battery plant b in period t , for each $b \in B, t \in \mathcal{T}$.
q_{belt}	Transportation capacity of 3PL provider l from b to e in period t , for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
US_{okt}	Supply capacity of supplier o for package k in period t , for each $o \in O, k \in K_o, t \in \mathcal{T}$.
$\xi_{et\varsigma}$	Demand at car manufacturing plant e in period t under scenario ς , for each $e \in E, t \in \mathcal{T}, \varsigma \in \Xi$.
A_{eokt}	Binary parameter: 1 if package k from supplier o serves plant e in period t ; 0 otherwise, for each $e \in E, o \in O, k \in K_o, t \in \mathcal{T}$.
y_{bt}	Binary variable: 1 if battery plant b is selected in period t ; 0 otherwise, for each $b \in B, t \in \mathcal{T}$.
x_{belt}	Binary variable: 1 if 3PL provider l is selected for route b - e in period t ; 0 otherwise, for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
p_{okt}	Binary variable: 1 if package k from supplier o is selected in period t ; 0 otherwise, for each $o \in O, k \in K_o, t \in \mathcal{T}$.
R_{bt}	Binary indicator variable: 1 if battery plant b has been selected before period t ; 0 otherwise, for each $t \in \mathcal{T}, b \in B$.
r_{belt}	Binary indicator variable: 1 if 3PL provider l for route b - e has been selected before period t ; 0 otherwise, for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}$.
γ_{ot}	Binary indicator variable: 1 if supplier o has been utilized before period t ; 0 otherwise, for each $o \in O, t \in \mathcal{T}$.
$sz_{bt\varsigma}$	Production quantity at battery plant b in period t under scenario ς , for each $b \in B, t \in \mathcal{T}, \varsigma \in \Xi$.
$z_{belt\varsigma}$	Quantity transported via 3PL provider l from b to e in period t under scenario ς , for each $b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}, \varsigma \in \Xi$.
$\phi_{et\varsigma}$	Unmet demand (shortage) quantity at car manufacturing plant e in period t under scenario ς , for each $e \in E, t \in \mathcal{T}, \varsigma \in \Xi$.
$\varpi_{okt\varsigma}$	Quantity procured in package k from supplier o in period t under scenario ς , for each $o \in O, k \in K_o, t \in \mathcal{T}, \varsigma \in \Xi$.

3.2. Model Formulation

Based on the defined notations, a multi-period, two-stage stochastic programming framework is constructed. During the initial phase, "here-and-now" decisions are finalized before stochastic demand manifests to manage multi-period capital expenditures and fixed costs. Specifically, decisions concerning battery plant location and activation, the selection of transportation routes and 3PL providers, and the procurement of external logistics service bundles via a winner determination mechanism are finalized, thereby establishing the fundamental topological structure of the supply chain. Subsequently, operational decisions are executed in the second stage under various stochastic demand scenarios to minimize multi-period operational and penalty costs. The initial stage of the model is mathematically structured as:

$$\begin{aligned} \text{Min} \quad & \sum_{t \in \mathcal{T}} \sum_{b \in B} H_{bt} y_{bt} (1 - R_{bt}) + \sum_{t \in \mathcal{T}} \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} h_{belt} x_{belt} (1 - r_{belt}) \\ & + \sum_{t \in \mathcal{T}} \sum_{o \in O} \sum_{k \in K_o} v_{ot} p_{okt} (1 - \gamma_{ot}) + E_{\xi} [f(\mathbf{y}, \mathbf{x}, \mathbf{p}, \mathbf{R}, \xi)] \end{aligned} \quad (\text{A1})$$

s.t.

$$\sum_{k \in K_o} p_{okt} \leq 1, \quad \text{for each } t \in \mathcal{T}, o \in O \quad (1)$$

$$N_{\min}^t \leq \sum_{o \in O} \sum_{k \in K_o} p_{okt} \leq N_{\max}^t, \quad \text{for each } t \in \mathcal{T} \quad (2)$$

$$\max \left\{ \max_{\substack{b \in B, e \in E \\ l \in L_{be}}} \{ \tau_{belt} x_{belt} \}, \max_{\substack{e \in E, o \in O \\ k \in K_o}} \{ A_{eokt} p_{okt} v_{eokt} \} \right\} \leq \kappa_t, \quad \text{for each } t \in \mathcal{T} \quad (3)$$

$$y_{bt} \leq 1, \quad \text{for each } b \in B, t \in \mathcal{T} \quad (4)$$

$$x_{belt} \leq 1, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \quad (5)$$

$$R_{bt} = \begin{cases} 1, & \sum_{v=1}^{t-1} y_{bv} > 0, \quad \text{for each } b \in B, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} y_{bv} = 0, \quad \text{for each } b \in B, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } b \in B, t = 1 \end{cases} \quad (6)$$

$$r_{belt} = \begin{cases} 1, & \sum_{v=1}^{t-1} x_{bekv} > 0, \quad \text{for each } b \in B, e \in E, k \in K_{be}, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} x_{bekv} = 0, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } b \in B, e \in E, k \in K_{be}, t = 1 \end{cases} \quad (7)$$

$$\gamma_{ot} = \begin{cases} 1, & \sum_{v=1}^{t-1} \sum_{k \in K_o} p_{okv} > 0, \quad \text{for each } o \in O, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} \sum_{k \in K_o} p_{okv} = 0, \quad \text{for each } o \in O, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } o \in O, t = 1 \end{cases} \quad (8)$$

$$p_{okt} \in \{0, 1\}, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T} \quad (9)$$

$$y_{bt} \in \{0, 1\}, \quad \text{for each } b \in B, t \in \mathcal{T} \quad (10)$$

$$x_{belt} \in \{0, 1\}, \quad \text{for each } b \in B, e \in E, l \in L_e, t \in \mathcal{T} \quad (11)$$

The expected operating cost $E_{\xi} [f(\mathbf{y}, \mathbf{x}, \mathbf{p}, \mathbf{R}, \xi)]$ is defined as follows:

$$\min \sum_{t \in \mathcal{T}} \int_{\xi \in \Xi} \left(\sum_{b \in B} s c_{bt} s z_{bt}(\xi) + \sum_{b \in B} \sum_{e \in E} \sum_{l \in K_{be}} k c_{belt} z_{belt}(\xi) \right. \\ \left. + \sum_{e \in E} p_{et} \phi_{et}(\xi) + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \varpi_{okt}(\xi) \right) \Phi(\xi) d\xi \quad (\text{A2})$$

s.t.

$$sz_{bt\zeta} \leq \Lambda y_{bt}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (12)$$

$$\sum_{e \in E} \sum_{l \in L_{be}} z_{belt\zeta} \leq sz_{bt\zeta}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (13)$$

$$sz_{bt\zeta} \leq Sq_{bt}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (14)$$

$$z_{belt\zeta} \leq x_{belt} q_{belt}, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}, \zeta \in \Xi \quad (15)$$

$$\sum_{b \in B} \sum_{l \in L_{be}} z_{belt\zeta} + \sum_{o \in O} \sum_{k \in K_o} A_{eokt} \varpi_{okt\zeta} = \xi_{et\zeta} - \phi_{et\zeta}, \quad \text{for each } e \in E, t \in \mathcal{T}, \zeta \in \Xi \quad (16)$$

$$\varpi_{okt\zeta} \leq p_{okt} U S_{okt}, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T}, \zeta \in \Xi \quad (17)$$

$$z_{belt\zeta} \geq 0, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}, \zeta \in \Xi \quad (18)$$

$$\phi_{et\zeta} \geq 0, \quad \text{for each } e \in E, t \in \mathcal{T}, \zeta \in \Xi \quad (19)$$

$$sz_{bt\zeta} \geq 0, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (20)$$

$$\varpi_{okt\zeta} \geq 0, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T}, \zeta \in \Xi \quad (21)$$

Equation (A1) defines the minimization of the anticipated aggregate expenditure by incorporating both multi-period and multi-scenario considerations. This function is composed of two distinct cost categories: the first category encompasses the fixed costs of various facilities in the first stage, objectively reflecting the continuous operational effects of facilities and routes over a long horizon; in contrast, the second category represents the expected operational costs across all demand scenarios in the second stage, as delineated in objective function (A2). Supplier parcel selection is governed by Constraints (1)–(2). Specifically, Constraint (1) stipulates the uniqueness of the selected parcel provided by a supplier within a single period, while Constraint (2) defines the upper and lower bounds for procurement quantities. Maximum transportation times involving self-built factories and external suppliers are restricted by Constraint (3). The operational status of network facilities and transportation routes is defined through Constraints (6)–(8). Furthermore, the binary attributes and integer restrictions for all decision variables are specified in Constraints (4)–(5) and (9)–(11). Factory capacity limitations are imposed by Constraints (12)–(15). Constraint (12) limits production capacity based on factory inventory capacity Λ , whereas Constraints (13)–(15) ensure that transportation flows comply with path selection variables and physical capacity limits. Network supply-demand equilibrium is established via Constraint (16), which balances stochastic demand $\xi_{et\zeta}$ by integrating production quantities, procurement volumes, and unmet demand $\phi_{et\zeta}$. Additionally, Constraint (17) mandates that the actual procurement quantity of parcels from external suppliers must not exceed the corresponding supply upper limit. Finally, the non-negativity properties of relevant decision variables are enforced by Constraints (18)–(21).

To facilitate an effective solution, the two-stage stochastic framework is recast to take the form of a mixed-integer nonlinear programming (MINLP) structure. Specifically, the anticipated operating cost term within the objective function is substituted with the sample mean over a discrete scenario set Ξ . Consequently, First-stage deterministic costs together with the weighted average of second-stage operational costs across all stochastic scenarios are jointly minimized. Furthermore, the first-stage strategic decision constraints are integrated with the second-stage operational decision constraints corresponding to each specific demand scenario to formulate an equivalent model. The specific form of the reformulated MINLP model, denoted as P , is presented as follows:

$$\begin{aligned} \min \quad & \sum_{t \in \mathcal{T}} \sum_{b \in B} H_{bt} y_{bt} (1 - R_{bt}) + \sum_{t \in \mathcal{T}} \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} h_{belt} x_{belt} (1 - r_{belt}) + \sum_{t \in \mathcal{T}} \sum_{o \in O} \sum_{k \in K_o} v_{ot} p_{okt} (1 - \gamma_{ot}) \\ & + \frac{1}{N} \sum_{\zeta=1}^N \left[\sum_{b \in B} s c_{bt} s z_{bt\zeta} + \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} k c_{belt} z_{belt\zeta} + \sum_{e \in E} p_{et} \phi_{et\zeta} + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \varpi_{okt\zeta} \right] \end{aligned} \quad (P)$$

s.t.

$$\sum_{k \in K_o} p_{okt} \leq 1, \quad \text{for each } t \in \mathcal{T}, o \in O \quad (22)$$

$$N_{min}^t \leq \sum_{o \in O} \sum_{k \in K_o} p_{okt} \leq N_{max}^t, \quad \text{for each } t \in \mathcal{T} \quad (23)$$

$$\max \left\{ \max_{\substack{b \in B, e \in E \\ l \in L_{be}}} \{ \tau_{belt} x_{belt} \}, \max_{\substack{e \in E, o \in O \\ k \in K_o}} \{ A_{eokt} p_{okt} v_{eokt} \} \right\} \leq \kappa_t, \quad \text{for each } t \in \mathcal{T} \quad (24)$$

$$y_{bt} \leq 1, \quad \text{for each } b \in B, t \in \mathcal{T} \quad (25)$$

$$x_{belt} \leq 1, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \quad (26)$$

$$sz_{bt\zeta} \leq \Lambda y_{bt}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (27)$$

$$\sum_{e \in E} \sum_{l \in L_{be}} z_{belt\zeta} \leq sz_{bt\zeta}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (28)$$

$$sz_{bt\zeta} \leq Sq_{bt}, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (29)$$

$$z_{belt\zeta} \leq x_{belt} q_{belt}, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}, \zeta \in \Xi \quad (30)$$

$$\sum_{b \in B} \sum_{l \in L_{be}} z_{belt\zeta} + \sum_{o \in O} \sum_{k \in K_o} A_{eokt} \varpi_{okt\zeta} = \xi_{et\zeta} - \phi_{et\zeta}, \quad \text{for each } e \in E, t \in \mathcal{T}, \zeta \in \Xi \quad (31)$$

$$\varpi_{okt\zeta} \leq p_{okt} U S_{okt}, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T}, \zeta \in \Xi \quad (32)$$

$$z_{belt\zeta} \geq 0, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T}, \zeta \in \Xi \quad (33)$$

$$\phi_{et\zeta} \geq 0, \quad \text{for each } e \in E, t \in \mathcal{T}, \zeta \in \Xi \quad (34)$$

$$sz_{bt\zeta} \geq 0, \quad \text{for each } b \in B, t \in \mathcal{T}, \zeta \in \Xi \quad (35)$$

$$\varpi_{okt\zeta} \geq 0, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T}, \zeta \in \Xi \quad (36)$$

$$R_{bt} = \begin{cases} 1, & \sum_{v=1}^{t-1} y_{bv} > 0, \quad \text{for each } b \in B, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} y_{bv} = 0, \quad \text{for each } b \in B, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } b \in B, t = 1 \end{cases} \quad (37)$$

$$r_{belt} = \begin{cases} 1, & \sum_{v=1}^{t-1} x_{bekv} > 0, \quad \text{for each } b \in B, e \in E, k \in K_{be}, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} x_{bekv} = 0, \quad \text{for each } b \in B, e \in E, k \in K_{be}, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } b \in B, e \in E, l \in L_{be}, t = 1 \end{cases} \quad (38)$$

$$\gamma_{ot} = \begin{cases} 1, & \sum_{v=1}^{t-1} \sum_{k \in K_o} p_{okv} > 0, \quad \text{for each } o \in O, t \in \mathcal{T} \setminus \{1\} \\ 0, & \sum_{v=1}^{t-1} \sum_{k \in K_o} p_{okv} = 0, \quad \text{for each } o \in O, t \in \mathcal{T} \setminus \{1\} \\ 0, & \text{for each } o \in O, t = 1 \end{cases} \quad (39)$$

$$p_{okt} \in \{0, 1\}, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T} \quad (40)$$

$$y_{bt} \in \{0, 1\}, \quad \text{for each } b \in B, t \in \mathcal{T} \quad (41)$$

$$x_{belt} \in \{0, 1\}, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \quad (42)$$

Despite the deterministic nature of Model P, high structural complexity and a significant computational burden are imposed by its MINLP formulation. Consequently, global optimal solutions are often not attained by general-purpose solvers within a reasonable time limit, particularly as the scale of instances increases. To address the computational intractability, a computationally tractable formulation is derived by further linearizing the nonlinear terms in Model P. Specifically, auxiliary binary decision variables u_{bt} , w_{belt} , and π_{okt} are introduced to substitute the nonlinear terms $y_{bt}(1 - R_{bt})$, $x_{belt}(1 - r_{belt})$, and $p_{okt}(1 - \gamma_{ot})$, respectively. Whereby, the associated nonlinear objective terms are equivalently transformed into linear forms, and the logical constraints are reformulated as follows:

$$u_{bt} \geq y_{bt} - R_{bt}, \quad \text{for each } b \in B, t \in \mathcal{T} \quad (43)$$

$$w_{belt} \geq x_{belt} - r_{belt}, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \quad (44)$$

$$\pi_{okt} \geq p_{okt} - \gamma_{ot}, \quad \text{for each } o \in O, k \in K_o, t \in \mathcal{T} \quad (45)$$

$$R_{bt} \leq \sum_{v=1}^{t-1} y_{bv}, \quad \text{for each } b \in B, t \in \mathcal{T} \setminus \{1\} \quad (46)$$

$$r_{belt} \leq \sum_{v=1}^{t-1} x_{belv}, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \setminus \{1\} \quad (47)$$

$$\gamma_{ot} \leq \sum_{v=1}^{t-1} \sum_{k \in K_o} p_{okv}, \quad \text{for each } o \in O, t \in \mathcal{T} \setminus \{1\} \quad (48)$$

Furthermore, nonlinear Constraint (3) involving the maximum operator is decomposed into the following set of linear constraints:

$$\tau_{belt} x_{belt} \leq \kappa_t, \quad \text{for each } b \in B, e \in E, l \in L_{be}, t \in \mathcal{T} \quad (49)$$

$$A_{eokt} p_{okt} v_{eokt} \leq \kappa_t, \quad \text{for each } e \in E, o \in O, k \in K_o, t \in \mathcal{T} \quad (50)$$

$$u_{bt}, w_{belt}, \pi_{okt} \geq 0, \quad \text{for each } b \in B, e \in E, l \in L_{be}, o \in O, k \in K_o, t \in \mathcal{T} \quad (51)$$

Through this process, the original formulation is transformed into a MILP model, denoted as (P-L). Despite the simplification achieved via linearization, significant complexity is retained in the problem due to the ‘‘Curse of Dimensionality.’’ Specifically, as the scale of the stochastic scenario set Ξ expands, a linear expansion in the number of variables and constraints leads to an exponential rise in solution time. Therefore, a SRDBA algorithm framework is developed in the subsequent section to efficiently solve Model (P-L).

4. A Scenario Relaxation and Decomposition-based Approximation Algorithm

In this section, an SRDBA algorithm is developed through the integration of SAA, LHS methods, dual decomposition, and Lagrangian relaxation algorithms. When samples are processed by the traditional SAA method, Monte Carlo simulation tends to be utilized for sample generation, and an optimal sample quantity is determined via large-scale numerical experiments. Computational efficiency may be reduced and the stability of the results may be compromised by the random selection of the sample size. According to Lo et al. (2013), equivalent or even superior effects can be achieved by the LHS method with a sample size only one-fifth of that required by Monte Carlo sampling. In this study, a strategy combining the LHS method and the chi-square test is adopted to substitute traditional repeated testing and identify an appropriate sample size. In this way, the structure of the SAA algorithm is reformulated, and Lagrangian relaxation along with dual decomposition methods are incorporated for efficiently obtaining a lower bound of the problem. Specifically, the utilization of LHS and Chi-square tests to enhance the solution scheme of the traditional SAA algorithm is elaborated in Section 4.1, thereby providing a point estimate of the upper bound for Model (P1). Subsequently, a method for the point estimation of the lower bound for Model (P1) is presented in Section 4.2.

4.1. Upper Bound Calculation

In this section, a method for estimating the upper bound of Model (A1) is developed. Initially, the optimal sample size is determined based on LHS and the Chi-square test. To fully verify the superior performance of the LHS method, a small-scale case study is presented in Table 2. In this case, sampling quality is assessed using a chi-square test, while, the deviation of the empirical distribution from the true distribution is quantified.

Table 2
Chi-square test results corresponding to different sample sizes.

N	10	20	30	50	100
$E[\mu - \bar{\mu}]$	0.341	0.133	0.074	0.033	0.011
$E[\sigma^2 - \bar{\sigma}^2]$	11.219	3.768	2.267	0.810	0.282
χ^2	24.16965	0	0	0	0
P -value	1	1	1	1	1

In Table 2, the mean and variance of the true random variables are denoted by μ and σ^2 , while the corresponding sample mean and variance are represented by $\bar{\mu}$ and $\bar{\sigma}^2$. As indicated by the tabulated results, when the sample size N is set to 20, the χ^2 and P values are 0 and 1, respectively. This demonstrates that the true distribution of the random variables is effectively approximated by the LHS method, even when a relatively small sample size is employed.

An optimal sample size $\hat{N}$ is assumed to be determined via LHS and the Chi-square test. Based on $\hat{N}$, the linearized Model P is solved, with the corresponding optimal solution denoted as y^*, x^*, p^* . Subsequently, a large sample of size $\hat{N}$ (where $\hat{N} \gg \hat{N}$) is introduced. Upon substituting this large sample and the solution y^*, x^*, p^* into the linearized Model P , the formulation is reduced to the following linear form:

$$\begin{aligned} \min_{\varpi, z, \phi, sz} \quad & \text{FCost} + \frac{1}{\hat{N}} \sum_{t \in \mathcal{T}} \sum_{\zeta=1}^{\hat{N}} \left[\sum_{b \in B} sc_{bt} sz_{bt\zeta} + \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} kc_{belt} z_{belt\zeta} \right. \\ & \left. + \sum_{e \in E} p_{et} \phi_{et\zeta} + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \varpi_{okt\zeta} \right] \\ \text{s.t.} \quad & (22)-(42) \end{aligned} \quad (P1)$$

where

$$\text{FCost} = \sum_{t \in \mathcal{T}} \sum_{b \in B} H_{bt} y_{bt} (1 - R_{bt}) + \sum_{t \in \mathcal{T}} \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} h_{belt} x_{belt} (1 - r_{belt}) + \sum_{t \in \mathcal{T}} \sum_{o \in O} \sum_{k \in K_o} v_{ot} p_{okt} (1 - \gamma_{ot})$$

represents a fixed constant. Notably, Model (P1) admits a decomposition that yields several independent components. The corresponding formulation for index ζ is presented below:

$$\begin{aligned} \min_{\varpi, z, \phi, sz} \quad & \text{Oper}_{\zeta} = \text{FCost} + \sum_{t \in \mathcal{T}} \left[\sum_{b \in B} sc_{bt} sz_{bt\zeta} + \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} kc_{belt} z_{belt\zeta} \right. \\ & \left. + \sum_{e \in E} p_{et} \phi_{et\zeta} + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \varpi_{okt\zeta} \right] \\ \text{s.t.} \quad & (22)-(42) \end{aligned} \quad (P1s)$$

By solving the corresponding (P1s) problem, a point estimate of an upper bound for P is obtained as

$$M_{\hat{N}}(y^*, x^*, p^*) = \text{FCost} + \frac{1}{\hat{N}} \sum_{\zeta=1}^{\hat{N}} \text{Oper}_{\zeta} \quad (52)$$

To determine the variance of the sample mean, $S_{\hat{N}}^2(y^*, x^*, p^*)$ is defined as the variance of the sample ($\text{FCost} + \text{Oper}_{\zeta}$) for $\zeta = 1 \dots \hat{N}$. Consequently, the variance of the sample mean is calculated via:

$$\sigma_{\hat{N}}^2 = \frac{1}{\hat{N}} S_{\hat{N}}^2(y^*, x^*, p^*) \quad (53)$$

4.2. Lower Bound Calculation

Sample sets, each of size N , are generated. To obtain the objective function values $\bar{M}_N^x$ ($x = 1, 2, \dots, \Gamma$), the linearized problem P is solved for each generated sample set. Consequently, the mean and sample variance of the objective function values are calculated via the following formulas:

$$\hat{M}_{N,\Gamma} = \frac{1}{\Gamma} \sum_{x=1}^{\Gamma} \bar{M}_N^x \quad (54)$$

$$\hat{\sigma}_{\bar{M}_{N,\Gamma}}^2 = \frac{1}{\Gamma(\Gamma-1)} \sum_{x=1}^{\Gamma} \left(\bar{M}_N^x - \hat{M}_{N,\Gamma} \right)^2 \quad (55)$$

Based on the SAA method developed by Kleywegt et al. (2002), $\widehat{M}_{N,\Gamma}$ is identified as a lower bound point estimate for problem A1. Given that the linearized version of problem P constitutes a MILP model, such lower bound estimates for small-to-medium-scale problems are typically obtained via the CPLEX solver. With the expansion of problem scale or the increase in the number of scenarios, if the linearized P with sample size N cannot be solved by commercial software such as CPLEX within a limited time, an algorithmic approach capable of efficiently obtaining the lower bound is provided in this paper based on dual decomposition and Lagrangian relaxation algorithms.

The computational intractability of the linearized model P is primarily attributed to the design of decision variables $\mathbf{y}, \mathbf{x}, \mathbf{p}$ being independent of the specific scenario ς . To address this challenge, the variables $\mathbf{y}, \mathbf{x}, \mathbf{p}$ are decomposed into the following form:

$$(\mathbf{y}_\varsigma, \mathbf{x}_\varsigma, \mathbf{p}_\varsigma), \quad \varsigma = 1, \dots, N \quad (56)$$

To ensure equivalence between the post-decomposition problem and the original formulation, the following constraints are incorporated into the model:

$$(\mathbf{y}_\varsigma, \mathbf{x}_\varsigma, \mathbf{p}_\varsigma) = (\mathbf{y}_\varrho, \mathbf{x}_\varrho, \mathbf{p}_\varrho), \quad \forall \varsigma, \varrho \in \Xi, \varsigma \neq \varrho \quad (57)$$

Consequently, model P is reformulated into the following form:

$$\begin{aligned} \min_{\substack{\mathbf{y}, \mathbf{x}, \mathbf{p}, \boldsymbol{\omega}, \\ \mathbf{z}, \boldsymbol{\phi}, \mathbf{s}, \mathbf{z}}} \quad & \sum_{t \in \mathcal{T}} \sum_{b \in B} H_{bt} y_{bt} (1 - R_{bt}) + \sum_{t \in \mathcal{T}} \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} h_{belt} x_{belt} (1 - r_{belt}) \\ & + \sum_{t \in \mathcal{T}} \sum_{o \in O} \sum_{k \in K_o} v_{ot} p_{okt} (1 - \gamma_{ot}) \\ & + \frac{1}{N} \sum_{t \in \mathcal{T}} \sum_{\varsigma \in \Xi} \left[\sum_{b \in B} s c_{bt} s z_{bt\varsigma} + \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} k c_{belt} z_{belt\varsigma} \right. \\ & \left. + \sum_{e \in E} p_{et} \phi_{et\varsigma} + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \omega_{okt\varsigma} \right] \end{aligned} \quad (P2)$$

s.t.

$$(\mathbf{y}_\varsigma, \mathbf{x}_\varsigma, \mathbf{p}_\varsigma) = (\mathbf{y}_\varrho, \mathbf{x}_\varrho, \mathbf{p}_\varrho), \quad \forall \varsigma, \varrho \in \Xi, \varsigma \neq \varrho \quad (58)$$

Constraint (58) functions as the non-anticipativity constraint, which introduces computational complexity to the model solution. Drawing upon PHA, auxiliary variables $\mathbf{y}, \mathbf{x}, \mathbf{p}$ are introduced to reformulate the aforementioned constraint as:

$$\mathbf{y}, \mathbf{x}, \mathbf{p} = \mathbf{y}_\varsigma, \mathbf{x}_\varsigma, \mathbf{p}_\varsigma, \quad \forall \varsigma \in \Xi, \quad \mathbf{y}, \mathbf{x} \in \{0, 1\} \quad (59)$$

With λ defined as the Lagrangian multiplier vector, Equation (59) is relaxed to establish the LR model, which is formulated as follows:

$$\begin{aligned} LR(\lambda) = \min_{\substack{\mathbf{y}, \mathbf{x}, \mathbf{p}, \boldsymbol{\omega}, \\ \mathbf{z}, \boldsymbol{\phi}, \mathbf{s}, \mathbf{z}}} \quad & \frac{1}{N} \sum_{\varsigma=1}^N \left\{ \sum_{t \in \mathcal{T}} \sum_{b \in B} H_{bt} y_{bt} (1 - R_{bt}) \right. \\ & + \sum_{t \in \mathcal{T}} \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} h_{belt} x_{belt} (1 - r_{belt}) \\ & + \sum_{t \in \mathcal{T}} \sum_{o \in O} \sum_{k \in K_o} v_{ot} p_{okt} (1 - \gamma_{ot}) \\ & + \sum_{t \in \mathcal{T}} \left[\sum_{b \in B} s c_{bt} s z_{bt\varsigma} + \sum_{b \in B} \sum_{e \in E} \sum_{l \in L_{be}} k c_{belt} z_{belt\varsigma} \right. \\ & \left. + \sum_{e \in E} p_{et} \phi_{et\varsigma} + \sum_{o \in O} \sum_{k \in K_o} b_{okt} \omega_{okt\varsigma} \right] \\ & \left. + \sum_{\varsigma \in \Xi} \lambda^T H_\varsigma(\mathbf{y}_\varsigma, \mathbf{x}_\varsigma, \mathbf{p}_\varsigma) \right\} \end{aligned} \quad (LR)$$

The Lagrangian relaxation formulation admits decomposition into several independent components, thereby improving computational efficiency. Correspondingly, a lower bound for the Lagrangian relaxation problem can be obtained by solving the following dual model.

$$LD(\lambda) = \max_{\lambda} LR(\lambda) \quad (LD)$$

Owing to the non-differentiability of the objective function, the Lagrangian dual model can be characterized as a concave maximization problem. To address this specific property, a subgradient method is employed for the solution process. The detailed calculation process is described as follows:

Based on Yin et al. (2024), it can be derived that $\sum_{\zeta=1}^N \mathbf{H}_{\zeta}(\mathbf{y}_{\zeta}^*, \mathbf{x}_{\zeta}^*, \mathbf{p}_{\zeta}^*)$ is a subgradient of problem (LD). Based on this subgradient information, the Lagrange multipliers can be updated according to the following formula:

$$\lambda^{k+1} = \lambda^k + s^k \sum_{\zeta=1}^N \mathbf{H}_{\zeta}(\mathbf{y}_{\zeta}^*, \mathbf{x}_{\zeta}^*, \mathbf{p}_{\zeta}^*)^k. \quad (60)$$

The step size s^k is determined using the adaptive step-size updating rule proposed by Yin et al. (2024), $s^k = \frac{\alpha^k(\beta^k - LR(\lambda^k))}{\|\sum_{\zeta=1}^N \mathbf{H}_{\zeta}(\mathbf{y}_{\zeta}^*, \mathbf{x}_{\zeta}^*, \mathbf{p}_{\zeta}^*)^k\|^2}$, where $\alpha^k = \varepsilon'(k)$ and $\beta^k = (1 + \varepsilon(k))\Phi^*$. Here, Φ^* denotes the best objective function value currently obtained, and $\varepsilon'(k)$ and $\varepsilon(k)$ are positive numbers that gradually decrease as the algorithm iterates. In this study, $\varepsilon'(k)$ and $\varepsilon(k)$ are set as linearly decreasing functions with respect to the iteration index k .

The stopping criterion of the algorithm is defined as

$$\left| \frac{LR(\lambda^{k+1}) - LR(\lambda^k)}{LR(\lambda^k)} \right| \leq \varepsilon, \quad (61)$$

where ε denotes a prescribed tolerance. The procedure stops once this threshold is met; if not, the multipliers are adjusted and the objective value is recalculated accordingly. From duality theory, the solution to (LD) yields a valid lower bound on the original problem.

4.3. GAP Calculation Formula

To evaluate the quality of the upper and lower bound solutions, the following computational formula is proposed to calculate the optimality gap:

$$\text{gap}(\mathbf{y}^*, \mathbf{x}^*, \mathbf{p}^*) = M_{\hat{N}}(\mathbf{y}^*, \mathbf{x}^*, \mathbf{p}^*) - \widehat{M}_{N,\Gamma} \quad (62)$$

$$\sigma_{\text{gap}}^2 = \sigma_{\hat{N}}^2 + \widehat{\sigma}_{\widehat{M}_{N,\Gamma}}^2 \quad (63)$$

Specifically, the optimal gap at the $1 - \alpha$ confidence level is calculated via the following formula:

$$M_{\hat{N}}(\mathbf{y}^*, \mathbf{x}^*, \mathbf{p}^*) - \widehat{M}_{N,\Gamma} \pm x_{\alpha} \sqrt{\sigma_{\hat{N}}^2 + \widehat{\sigma}_{\widehat{M}_{N,\Gamma}}^2} \quad (64)$$

Here, $x_{\alpha} = \Lambda^{-1}(1 - \alpha)$, and $\Lambda(x)$ is the standard normal cumulative distribution function.

5. Numerical Experiments

In this section, the sources of fundamental data are detailed, and experimental analyses are conducted on the algorithm and critical parameters. All numerical analyses are executed on a desktop system configured with a 2.11 GHz processor and 16 GB of RAM. The section is organized as follows: first, the sources of fundamental data values are elaborated in Section 5.1; subsequently, the solution performance and result quality of the algorithm are evaluated in Section 5.2. A comprehensive sensitivity analysis is conducted in Section 5.3 to investigate the impacts of variations in key model parameters, and finally, the predictive and decision-making effectiveness of three forecasting models is analyzed in Section 5.4.

5.1. Basic Data

To verify the feasibility of the model, a case study is conducted based on public information from BYD Auto. The network topology is simplified based on the actual industrial layout of the company, and a representative set comprising 4 battery production plants and 10 automobile production plants is constructed, as illustrated in Fig. 4. Data regarding facility locations, fixed costs, and production capacities are derived from the public financial reports and official news releases of BYD. Other model parameters, including transportation costs, time constraints, and demand fluctuations, are generated via random sampling based on industry standards within reasonable estimation ranges. As shown in the table 3, the parameters are categorized by type, with specific values and ranges explicitly listed.

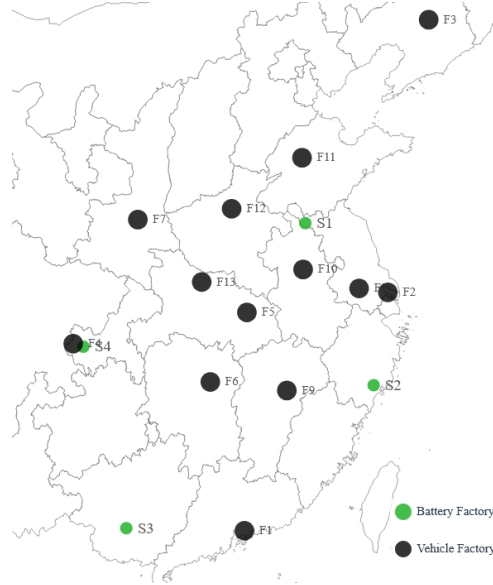


Figure 4: BYD factory network

5.2. Performance Analysis

This section evaluates the performance of the proposed algorithm based on numerical experimental results. The experimental settings encompass two scales of stochastic demand scenarios, comprising 20 and 1000 scenarios, respectively, and the SRDBA algorithm described previously is employed for the solution process. To ensure computational feasibility, a maximum runtime of 2 hours is imposed on each single run of the CPLEX solver, which is automatically terminated if this time limit is exceeded or if memory insufficiency occurs.

The expansion of problem size and scenario count makes the full-scenario model increasingly difficult to solve, thereby hindering the construction of a reliable benchmark for evaluating the SRDBA solution. Consequently, the Lower Bound (LB) and Upper Bound (UB) of the solution are introduced to evaluate the performance of the SRDBA algorithm. The results of the upper and lower bounds for each test instance are presented in Table 3.4. The data indicate that the Optimality Gap between the upper bound corresponding to the SRDBA solution and the lower bound for all instances is maintained within a small range. This demonstrates that the SRDBA method is able to produce a dependable lower estimate of the problem's true optimum. Simultaneously, the quality of the solution generated for evaluating the upper bound is close to optimal. Collectively, it is indicated that high-quality approximate solutions for this problem can be stably provided by the proposed algorithm.

⁶BYD. *BYD 2023 Annual Report*. Available at: <https://data.eastmoney.com/notices/detail/002594/AN202403261628203493.html> (accessed 1 January 2025).

⁷Tencent News. *New Planning of BYD Shenzhen Automobile Production Base*. Available at: <https://news.qq.com/rain/a/20250303A03P3S00> (accessed 1 January 2025).

⁸SF Express. *Time-efficiency and Price Query*. Available at: <https://www.sf-express.com/chn/sc/price-query> (accessed 1 January 2025).

⁹CATL. *CATL 2024 Annual Report*. Available at: https://www.catl.com/uploads/1/file/public/202503/20250317212347_bwwxumcf0.pdf (accessed 1 January 2025).

Table 3

Parameters description and values

Symbol	Description	Value / Range
<i>Network structure parameters</i>		
$\mathcal{T}$	Number of time periods	3
B	Number of EVB production plants	4
E	Number of EV manufacturing plants	13
O	Number of battery suppliers	10
L_{be}	Number of 3PL providers	2
Ξ	Number of scenarios	20
<i>Cost parameters (Unit: 10^4 CNY)</i>		
H_{bt}	Fixed cost of operating EVB production plant	180,000–500,000
v_{ot}	Maintenance cost for battery supplier	1,000–5,000
h_{belt}	Maintenance cost for 3PL provider	5–30
sc_{bt}	Unit production cost at EVB production plant	20–50
kc_{belt}	Unit transportation cost via 3PL provider	0.1–1
p_{et}	Penalty cost of unmet demand	50–200
b_{okt}	Procurement cost of package from battery supplier	50–500
<i>Capacity, Time & Demand parameters</i>		
Sq_{bt}	Production capacity of EVB plant (units)	3,000–40,000
US_{okt}	Supply capacity of battery supplier (units)	50–500
q_{belt}	Transportation capacity of 3PL provider (units)	1,500–2,500
$\xi_{et\zeta}$	Demand at EVB plant under scenario (units)	500–3,500
τ_{belt}	Transportation time via 3PL provider (h)	30–100
v_{eokt}	Trans. time from supplier to EVB plant (h)	40–120
κ_t	Maximum allowable time limit in period (h)	300–600

Note: Data are estimated based on the financial statements and industry reports of EYC and CATL from 2023 to 2024. Factory and production parameters, including H_{bt} and sc_{bt} , are derived from the BYD 2023 Annual Report⁶ and the strategic planning documents of the BYD Shenzhen automobile production base⁷. Similarly, transportation parameters such as v_{ot} and h_{belt} are determined based on SF Express time-efficiency quotations⁸, while external procurement parameters q_{belt} and b_{okt} are adopted from the CATL 2024 Annual Report⁹. Finally, the remaining parameters are reasonably estimated based on available data.

To accommodate the multi-period model data, the values in the table are presented as fluctuation intervals, while specific experimental values are determined by employing the sample average sampling method within these ranges.

Table 4

SRDBA GAP analysis

Penalty per period			Low bound		Upper bound		Gap		Gap (%)
t_1	t_2	t_3	Avg.	Std.	Avg.	Std.	Avg.	Std.	
39	41	46	2,308,839	144.22	2,310,771	3,743.70	1,932	3,746.47	0.3430
48	54	56	2,527,987	667.58	2,556,040	4,457.82	28,052	4,507.53	1.3796
62	65	63	2,616,483	965.64	2,626,628	4,292.42	10,145	4,399.70	0.6542
81	85	83	2,697,834	1,974.99	2,738,057	6,137.14	40,223	6,447.10	1.8457
112	145	122	2,764,006	1,815.59	2,795,951	4,542.94	31,944	4,892.31	1.4225

5.3. Parameter Analysis

A parameter sensitivity analysis is conducted in this chapter. Identifying demand variability, unit penalty costs, transportation time limits, and external procurement strategies as core factors, two independent sets of experiments are formulated. Specifically, these experiments are employed to systematically investigate the multidimensional impacts of the aforementioned factors on the Total Cost (TC), Availability Rate (AR), Penalty Cost (P), Transportation Cost (SC), and Construction Cost (JC) of the supply chain network.

In the first set of experiments, three core independent variables are established: the demand variability (CV), the activation of the winner determination problem mechanism (WDP), and the capacity level of self-built facilities (Sq). Through the combinatorial adjustment of these parameters, their comprehensive impacts on the TC , P , AR , as well as

specific cost components such as JC and SC , are thoroughly investigated. Regarding the second set of experiments, the service time window (T) and P are selected as key regulatory parameters. Consequently, the variations in TC and SC under varying service timeliness and reliability requirements are analyzed.

5.3.1. Analysis of the Impact of CV, WDP, and Sq on Results

To investigate the impacts of variations in CV , WDP , and Sq on the experimental outcomes, the demand volume d is formulated as $d = m + nu$, wherein the degree of demand fluctuation is characterized by n . The utilization of third-party suppliers and the WDP mechanism is denoted by binary indicators, where 1 indicates activation and 0 signifies deactivation. Specifically, the parameter values are set as follows: $m \in \{1412, 1550, 1283\}$, $n \in \{0.5, 3\}$, $u \in \{147, 208, 145\}$, and $Sq \in \{1000, 2000, 3500\}$. The computational results concerning TC , AR , JC , SC , and P are presented below Table 5, Fig. 6 and Fig. 5.

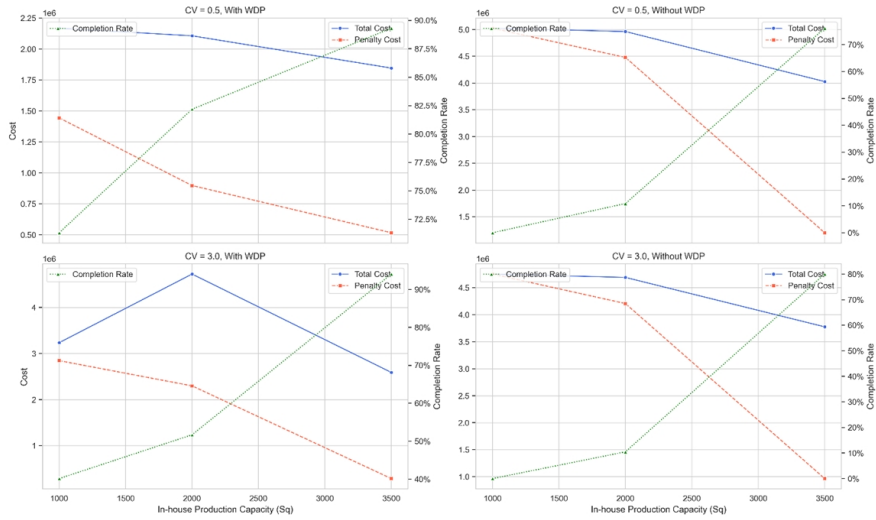


Figure 5: Data changes under different CV, WDP, and SQ

The complex interactions among the three critical parameters are revealed by the data presented in the table. Consequently, the following insights are derived:

- I. The fundamental improvement in supply chain network performance yielded by the activation of the WDP mechanism is demonstrated.
- II. Network performance can be continuously optimized by increasing the capacity of self-built factories; however, the marginal benefit is influenced by external procurement strategies.
- III. Strict challenges are posed to the cost and service level of the SC network by demand fluctuations, and differences under various strategic choices are significantly amplified.

Regarding experimental data, scenarios with disabled WDP are consistently associated with a sharp increase in TC and a substantial decline in demand AR . Particularly under conditions of severe Sq insufficiency, the inability to meet any demand is exhibited by systems with $WDP = 0$, where total costs are almost entirely constituted by substantial P , indicative of complete network failure. For instance, in a low-volatility environment, a 56.8% reduction in total costs and an increase in demand satisfaction rate to 71.31% are achieved by enabling WDP . The introduction of external suppliers and the adoption of the WDP mechanism are demonstrated as critical means for endowing the SC network with flexibility and resilience, whereby impacts from internal capacity shortages and demand fluctuations are effectively buffered by the 4PL through the integration of external resources. Regarding Sq , a downward trend in TC and a steady improvement in AR are exhibited across all scenarios as Sq is increased. The classic trade-off between JC and P is reflected; although construction costs are increased by capacity expansion, overall cost optimization is achieved via the consequent substantial reduction in penalty costs. A smoother and more economical nature of such improvements is observed under the enabled WDP mode. For instance, in a high-volatility environment,

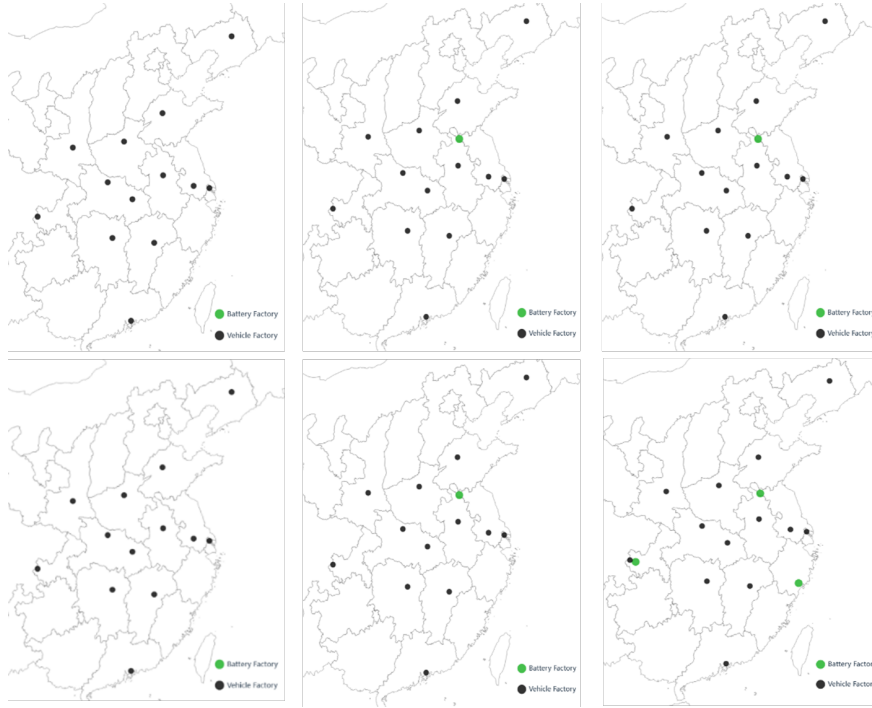


Figure 6: Partial cycle facility selection

a 45.1% reduction in TC and a 42.38% increase in demand satisfaction rate are attained when WDP is enabled alongside increased capacity, whereby significant advantages of synergies between internal and external resources are demonstrated. Conversely, a mere 19.5% reduction in TC is yielded by identical capacity increases when WDP is disabled, whereby limitations of sole reliance on internal expansion are highlighted. Regarding CV , general increases in P and TC alongside decreases in AR are observed under high-volatility environments across the compared datasets. For instance, a 40.3% increase in TC is yielded by demand fluctuations rising from 0.5 to 3.0 when WDP is enabled at a capacity of 3500. The critical value of flexible procurement strategies in addressing uncertainty is highlighted by the prominence of these effects under the combination of high volatility and disabled WDP . Peak demands are effectively absorbed by flexible external supplier networks, whereby the maintenance of high service levels and controllable costs in high-volatility markets is achieved; conversely, such performance is deemed difficult to attain via sole reliance on fixed internal capacity.

In summary, the establishment of a hybrid SC , governed by a 4PL and characterized by synergy between internal capacity and external supplier networks, is recognized as the optimal strategy for the attainment of the dual objectives of cost efficiency and service flexibility within uncertain market environments. Substantial costs are incurred by sole reliance on internal capacity expansion, and a high degree of vulnerability to demand fluctuations is demonstrated by such strategies. Critical strategic flexibility is afforded to network design via the implementation of the WDP mechanism.

5.3.2. Impact of Time Window T and Penalty P on Experimental Results

The time window T and supply reliability are regarded as core dimensions for evaluating the performance of a 4PL-driven EVBMS network. In the present analysis, particular emphasis is placed on examining how variations in two key parameters, namely the time window T and the penalty cost P , influence decision adjustments at the strategic, tactical, and operational levels. Within the logistics service procurement decisions of the proposed model, the existence of two alternative third-party logistics service providers on each transportation link is assumed. One provider offers a priority service characterized by high timeliness and higher charges, whereas the other provider offers a standard service characterized by lower cost and longer transportation time. Specifically, the time window T is defined within

Table 5

Experimental data under different CV, WDP, and SQ conditions

CV, WDP	Metric	Sq		
		1,000	2,000	3,500
0.5, 1	JC	31,415	341,560	341,660
	SC	0	1,427	3,170
	P	1,444,200	898,220	516,870
	AR	71.31	82.18	89.31
	TC	2,168,281	2,106,994	1,846,455
0.5, 0	JC	0	310,040	1,421,500
	SC	0	1,186	6,979
	P	5,024,200	4,478,200	1,202,200
	AR	0.00	10.87	76.13
	TC	5,024,200	4,962,586	4,027,274
3, 1	JC	31,415	341,560	1,073,100
	SC	0	949	6,791
	P	2,845,400	2,299,400	289,190
	AR	40.14	51.68	94.06
	TC	3,235,017	4,722,319	2,590,029
3, 0	JC	0	310,090	1,421,700
	SC	0	874	7,732
	P	4,752,800	4,206,800	964,880
	AR	0.00	10.47	80.02
	TC	4,752,800	4,690,877	3,775,449

the interval [40, 100], and the penalty cost P is selected from the set {55, 60, 65, 90, 120}. The results for TC , SC , and P are reported as follows in Table 6, Fig. 7, and Fig. 8.

Table 6Experimental data under different T , and P conditions

T	P	SC	TC
40	55	5,417	2,726,470
	60	9,934	2,853,611
	65	14,312	2,930,578
	90	14,202	2,933,779
	120	19,869	3,226,218
100	55	3,602	2,635,324
	60	6,663	2,754,213
	65	9,146	2,831,326
	90	9,299	3,244,516
	120	12,838	3,228,482

Based on the experimental results, the following managerial insights are derived:

- IV. Supply chain operating costs are systematically elevated by stricter time windows.
- V. The risk preference of the model is significantly altered by an increase in the unit shortage penalty, inducing a transition from a cost-oriented focus to a service-oriented strategy.
- VI. The impact of timeliness variations on the total cost is attenuated when customer requirements for supply reliability are maximized.

Regarding *IV*, both the SC and TC at $T = 40$ are observed to be significantly higher than the corresponding values at $T = 100$ across all scenarios of P . For example, when $P = 55$, compression of the time window from 100 to 40 is associated with a 50.4% increase in transportation cost and a corresponding 3.5% increase in total cost. Rational decision making in logistics service procurement within the model is reflected by this result, under which a

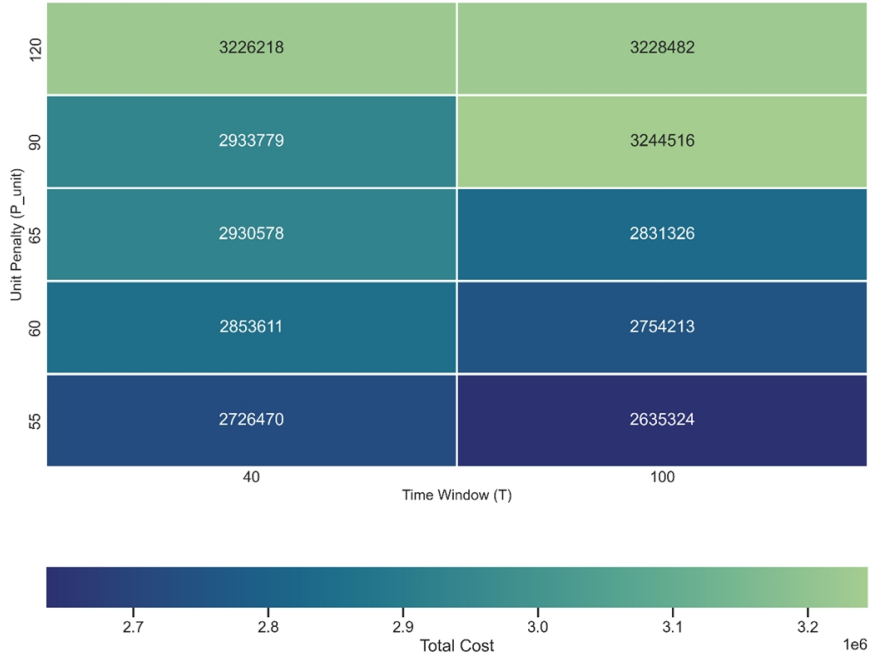


Figure 7: TC changes under different T and P

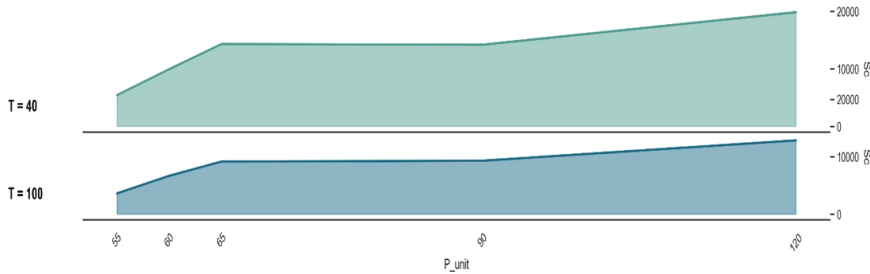


Figure 8: SC changes under different T and P

greater reliance on higher priced and faster priority logistics service providers is induced by more stringent timeliness requirements, and transportation costs are therefore directly increased. For V , a pronounced increasing trend in SC is observed as the value of P increases. Regarding example, under the scenario with $T = 100$, an increase of P from 55 to 120 is associated with a 256.4% increase in transportation cost. This indicates that, as the economic loss caused by shortages, represented by penalty costs, becomes increasingly high, greater investment in transportation activities is induced within the model, with a stronger preference given to priority logistics services characterized by high timeliness and reliability so as to minimize shortage risk, even when higher transportation costs are incurred. Through such strategic adjustment, the growth of total cost is kept relatively moderate, and the capability of the model to hedge against high penalty risk through optimization of the transportation mix is thereby demonstrated. Regarding VI , convergence of TC across the two time window scenarios is observed under extremely high penalty costs. Under such conditions, shortage avoidance is prioritized, and the fastest and most reliable transportation schemes are adopted regardless of whether the time window is 40 or 100, even at substantially higher cost. In this extreme setting, convergence of supply chain operating patterns is induced, and similarity in total cost is consequently exhibited.

The experiment verifies that variations in service level requirements are effectively captured and responded to by the model. Quantitative evaluation of cost implications under different service levels is enabled for decision makers according to heterogeneous customer requirements and market environments, thereby allowing an optimal balance

to be identified within the multidimensional decision space of cost, time, and reliability, and providing quantitative decision support for new energy vehicle enterprises in addressing timeliness challenges in EVB transportation.

5.4. Prediction Method Analysis

A novel multi-model comparative validation framework is established to systematically evaluate the potential impacts of distinct advanced forecasting techniques on supply chain decision-making. Grounded in the application of deep learning within the domain of time series forecasting, three representative models are selected, namely GRU, LSTM, and the Transformer model. These architectures are employed to comprehensively examine the capabilities of distinct models in capturing short-term dependencies, long-term trends, and complex patterns embedded within power battery demand sequences. To simulate the dynamic decision-making process characteristic of actual supply chain operations, a rolling forecasting mechanism defined as “predict-update-repredict” is adopted. Specifically, battery procurement data from a leading EV enterprise, spanning 72 months from 2018 to 2023, is utilized as the training set, while the actual demand for the first and second quarters of 2024 serves as the test set. The specific implementation procedure is detailed as follows.

- I. In the data preprocessing phase, Z-score normalization is applied to the raw data to eliminate dimensional differences.
- II. For model training, time series cross-validation is employed, with the sliding window length set to 6, the batch size to 32, and the initial learning rate to 0.001.
- III. An input sequence sliding window with a length of 12 months is established, wherein the initial input comprises the actual demand data from the final 12 months of the training set. In each forecasting cycle, demand forecasts for the subsequent three months are generated by the model. Subsequently, the predicted value for the first month of the three-month horizon is regarded as the “quasi-real value” and incorporated into the end of the input sequence, while the initial data point of the sequence is excluded, thereby causing the sliding window to shift forward by one month.

The Total Deviation (TD) is defined as:

$$TD = \frac{1}{3} \sum_{i=1}^3 (ED_i \cdot \omega_1 + DD_i \cdot \omega_2) \quad (65)$$

where ω_1 and ω_2 represent weight coefficients—specifically $\omega_1 = 0.8 \times 10^{-6}$ and $\omega_2 = 0.2 \times 10^{-12}$ —which are utilized to balance deviation values of differing magnitudes. $T1$, $T2$, and $T3$ denote the three periods, while Predicted Expectation (PE), Predicted Variance (PD), Expectation Deviation (ED), Variance Deviation (DD), and Total Deviation (TD) are defined accordingly. The specific data are presented as follows Table. 7 and Fig. 9.

Based on the evaluation results derived from the total deviation (TD) and the corresponding component indicators, the LSTM model is shown to exhibit the strongest forecasting performance, with predicted demand being closest to the actual demand observed in the test set in terms of both expectation and variance. However, a core principle of stochastic supply chain management lies in the recognition that statistical forecasting accuracy cannot be fully equated with effectiveness in practical decision making. The ultimate value of a forecasting model is not determined by isolated accuracy metrics, but should be evaluated according to the economic value generated after integration into an operational decision framework. An economic loss function is inherently constituted by the objective function of the MILP model established in the present study. Rather than merely penalizing the absolute magnitude of forecasting errors, forecasting quality is evaluated on the basis of a series of complex economic consequences associated with inventory decisions, penalty costs, and transportation strategy selection.

To validate the underlying hypothesis and to assess the decision effectiveness of each forecasting model from the perspective of economic benefit, the second stage of the experimental framework is conducted. In this stage, the demand forecasts generated by the GRU, LSTM, and Transformer models are respectively employed as demand-parameter inputs to the established MILP model. For each forecast sequence, the MILP model is solved to obtain the optimal supply chain network design and the corresponding total cost. By benchmarking these outcomes against the reference cost obtained by solving the same model using realized demand data, a more profound assessment is enabled regarding the true impact of alternative forecasting models on the ultimate quality of supply chain decisions; the detailed results are reported as follows Table. 8.

Table 7
Experimental data under different prediction models

Model	Metric	T_1	T_2	T_3
GRU	PE	1,985,770	2,388,407	2,512,122
	PD	2.78×10^{11}	2.07×10^{11}	2.98×10^{11}
	ED	642,367	48,407	472,122
	DD	7.07×10^{10}	2.05×10^{11}	1.37×10^{11}
	TD		0.3376	
LSTM	PE	1,377,464	1,793,096	2,079,185
	PD	1.39×10^{11}	6.02×10^{10}	2.78×10^{11}
	ED	34,062	546,904	39,185
	DD	2.10×10^{11}	5.75×10^{10}	1.57×10^{11}
	TD		0.1937	
Transform	PE	1,528,006	1,486,072	1,490,158
	PD	3.93×10^9	4.25×10^9	4.34×10^9
	ED	184,603	853,927	549,841
	DD	3.45×10^{11}	1.58×10^9	4.30×10^{11}
	TD		0.4754	

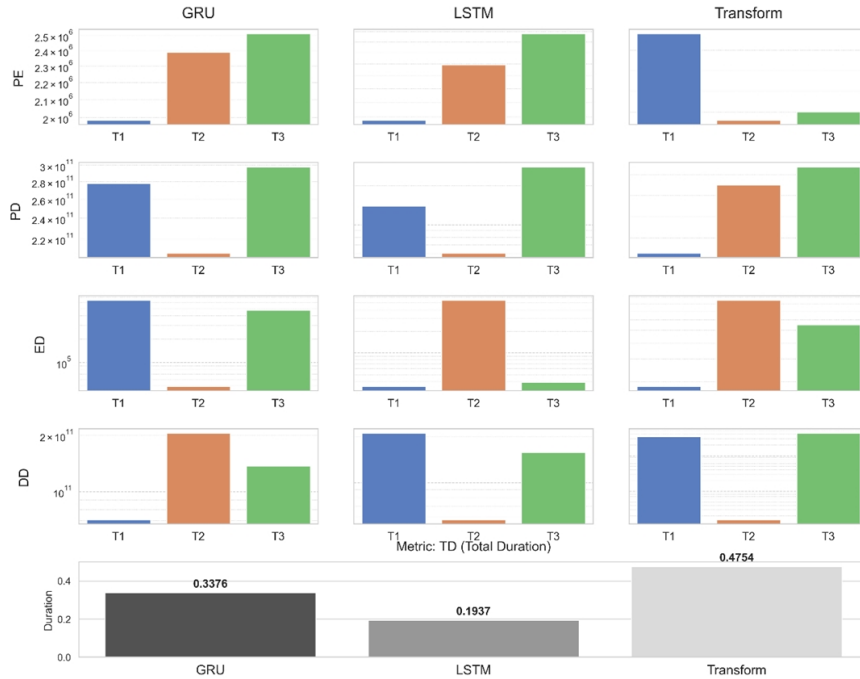


Figure 9: Comparative analysis of prediction models

The minimum total cost in the decision stage is not achieved by the LSTM model that is identified as optimal in the DF stage. In contrast, the GRU model with second-best forecasting accuracy drives an SCND solution whose total cost is closer to the benchmark cost obtained under realized demand. A misalignment between the evaluation criteria adopted for DF models and the optimization objective embedded in the decision model is therefore indicated. As a multi-period stochastic programming system, the MILP model developed in the present study induces a non-linear response of the objective function to forecasting errors. In particular, specific error patterns, such as severe underestimation of demand in critical periods, are associated with the activation of substantial penalty costs, whereby

Table 8

Experimental evaluation of different prediction models

Model	Real	Predicted	Metric	Composite
GRU		3,336,904	541,671	0.1937
LSTM	2,795,232	2,209,774	585,457	0.2094
Transform		1,683,585	1,111,646	0.3976

an impact on the total cost is exerted that is markedly greater than that implied by the corresponding statistical error magnitude. On this basis, the following inference is derived.

VII. A significant impact on decision quality is exerted by the distributional characteristics of forecasting errors.

In comparison with a single average-error metric, a more direct influence on the ultimate decision cost is exerted by the distributional structure of forecasting errors, particularly by the frequency and magnitude of extreme errors. Empirical evidence is provided that the forecast sequences generated by the GRU, although associated with a higher mean error than those generated by the LSTM, are characterized by a more effective avoidance of severe demand underestimation that would otherwise induce substantial penalty costs. Higher effectiveness within a cost-minimization decision model is thereby exhibited as a consequence of such risk-averse characteristics. Within the stochastic programming framework developed in the present study, risk-management mechanisms for uncertainty are internalized through the joint integration of capacity, temporal, and cost constraints. Consequently, an ideal forecasting input is required to provide information that facilitates a balance between expected cost and potential risk within the decision model. Superior satisfaction of such a requirement is exhibited by the GRU forecasts in the present case.

In summary, the experimental findings demonstrate that forecasting-model selection is required to be conducted with caution for multi-objective stochastic optimization problems such as supply chain production planning. Exclusive emphasis on the statistical accuracy of DF is liable to obscure the actual economic consequences transmitted to downstream decision-making. A collaborative “forecasting–decision” evaluation paradigm is adopted. Within such a paradigm, DF is provided as the baseline input to the stochastic programming model, whereas model effectiveness is ultimately assessed through the decision outcomes yielded under multi-factor dynamic optimization. The limitations of conventional one-dimensional evaluation of forecasting models are thereby avoided, and the selection of DF approaches is more closely aligned with enterprise operational objectives under complex business environments.

6. Conclusion

From the perspective of 4PL, a data-driven multi-period EVBMS design problem integrating a WD mechanism under uncertainty is investigated. To cope with the pronounced volatility in EV manufacturing demand, a closed loop decision framework integrating demand forecasting and network optimization is constructed, with supply chain flexibility and cost effectiveness enhanced through a balanced allocation between self owned capacity and third party procurement. In support of this framework, multi period demand fluctuation characteristics are forecast using three deep learning models, namely LSTM, GRU, and Transformer, and a data driven two stage multi period stochastic programming model is subsequently formulated. Specifically, strategic decisions involving facility location and third-party logistics selection are determined in the first stage, whereas operational costs under varying demand scenarios are optimized in the second stage. The model is subsequently reformulated as a MILP problem. For the purpose of ensuring the tractability of large scale instances, A Scenario Relaxation and Decomposition-based Approximation Algorithm, denoted as SRDBA, is proposed, in which LHS, SAA, and LR based on dual decomposition are integrated. Based on numerical experiments conducted using publicly available data from an EV manufacturing enterprise, the effectiveness of the proposed model and algorithm is verified, and the following key managerial insights are derived.

- I. **Synergistic effects between the WD mechanism and capacity planning.** Supply chain network performance is fundamentally improved by the introduction of the WD mechanism through the effective integration of third-party EVB suppliers. Although network performance is sustainably optimized by increasing self-built factory capacity, the marginal benefit is significantly influenced by external procurement strategies, thereby highlighting the necessity of a mixed procurement mode in balancing cost and flexibility.

- II. **Trade-off between service level requirements and risk preference.** Supply chain operational costs are systematically elevated by stricter time windows. Furthermore, the risk preference of the model is significantly altered by an increase in unit shortage penalties, driving decisions from a cost-oriented approach toward a service-oriented one. Notably, the impact of timeliness differences on total cost is attenuated when customer requirements for supply reliability reach an extreme level.
- III. **Evaluation strategy for the “prediction-decision” framework.** A significant non-linear relationship is observed between forecast accuracy and decision quality. The results indicate that decision quality is substantially influenced by the distributional characteristics of forecast errors. An evaluation mechanism for forecasting models oriented toward optimization driven decision outcomes rather than purely statistical accuracy is established, thereby ensuring that assessment criteria are more closely aligned with practical application value and preventing imbalances in which strong forecasting performance is accompanied by poor decision effectiveness.

The main contributions of this study are summarized as follows:

- I. A novel data-driven optimization framework is proposed. Distinct from traditional approaches, an evaluation mechanism for prediction models based on optimization results is designed to realize the dynamic feedback and closed-loop interaction between demand prediction and optimization decisions.
- II. The WD method is innovatively integrated into 4PLEVBMSC network design. Through this integration, third-party EVB suppliers are incorporated into 4PL architecture, thereby enhancing the flexibility and robustness of the network in accommodating market fluctuations.
- III. A novel data-driven two-stage stochastic multi-period model is formulated, and an efficient SRDBA algorithm is developed. Specifically, by integrating LHS with dual decomposition and Lagrangian relaxation, an effective computational framework is established to address complex stochastic network design problems.

Despite the aforementioned contributions, further extensions are permitted in several aspects in future research. First, while the primary focus of this study is placed on demand uncertainty, future investigations could be extended to accommodate other sources of uncertainty, such as transportation cost fluctuations, variations in raw material prices, or carbon emission policies. Second, the proposed decision-driven framework is applicable to other prediction-dependent supply chain design problems. Furthermore, to enhance the dynamic coupling capability between prediction and optimization within complex real-time decision-making environments, the integration of advanced technologies, such as deep reinforcement learning, is suggested.

Acknowledgements

Funding: This work was supported by the China Postdoctoral Science Foundation (2025MD784090); the Natural Science Foundation of Liaoning Province (2024-BS-227); the Social Science Foundation of Liaoning Province (L24CGL031); the “Xingliao Talent Program” Leading Talents Project of Liaoning Province (XLYC2402025); the National Local Joint Engineering Laboratory Project for Optimization of Petrochemical Process Operations and Energy Saving Technology of the Education Department of Liaoning Province (LJ232510148001); the Foundation of Talent Scientific Research Fund of LNPU (2023XJLL-004).

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